Train Coaches Both Ways

Category B Coding Assignment - Problem 4

Problem summary

Maintain a doubly linked list with head and tail references. Add coaches at the end, remove the first coach with a matching name, then print forward and backward.

Java solution

```
public class Main {
 static class Node {String name;Node prev,next;Node(String n){name=n;}}
 static class Train {Node head,tail;
  void addLast(String name){Node n=new Node(name);if(head==null)head=tail=n;else{tail.next=n;n.prev=tail;tail=n;
  boolean remove(String name){Node n=head;while(n!=null&&!n.name.equals(name))n=n.next;if(n==null)return false;
   if(n.prev!=null)n.prev.next=n.next;else head=n.next;
   if(n.next!=null)n.next.prev=n.prev;else tail=n.prev;return true;}
  void printForward(){System.out.print("Forward: ");for(Node n=head;n!=null;n=n.next){System.out.print(n.name);i
  void printBackward(){System.out.print("Backward: ");for(Node n=tail;n!=null;n=n.prev){System.out.print(n.name)
 }
 public static void main(String[] args){Train t=new Train();for(String s:new String[]{"Engine","A1","B1","B2","G
}
```

Sample output

```
Forward: Engine <-> A1 <-> B2 <-> Guard
Backward: Guard <-> B2 <-> A1 <-> Engine
```

Complexity

addLast: O(1) using tail. Finding the coach: O(n). Unlinking after finding: O(1). Forward/backward printing: O(n).